

How Well Does AI Understand Nepali?

A Multi-Task Benchmark of Large Language Models and Embedding Models on a Low-Resource Language

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Abstract—Modern language models are trained predominantly on English, yet they are increasingly deployed to speakers of low-resource languages. This paper presents a reproducible multi-task benchmark measuring how well current large language models (LLMs) and text embedding models handle Nepali, a Devanagari-script language spoken by more than 30 million people. Five tasks are evaluated: tokenizer fertility, paired reading comprehension, sentiment classification of native social media text, machine translation in both directions, and embedding-based passage retrieval. All evaluation data derives from openly licensed sources and every result is generated by scripts published with the raw model outputs. Three main findings emerge. First, tokenization imposes a direct cost penalty on Nepali users: on identical parallel content, the GPT-4 era cl100k tokenizer consumes 4.79 times more tokens for Nepali than for English, and popular open-model tokenizers consume 2.1 to 4.5 times more, whereas multilingual-first tokenizers approach parity. Second, comprehension degrades sharply with model scale and recency: on identical Belebele questions, Llama 3.1 8B falls from 84.0% accuracy in English to 40.5% in Nepali, while frontier-class models retain most of their ability, with Gemini 3.1 Flash Lite falling only from 97.0% to 89.5%. All within-model language gaps are statistically significant under McNemar’s exact test. Third, capability is asymmetric: models translate out of Nepali far better than into it, and even strong models reach only 51.9% to 67.1% accuracy on three-class sentiment classification of informal native Nepali text. On retrieval, modern multilingual embedding models perform well (BGE-M3 reaches 84.0% Recall@1), showing that Nepali retrieval-augmented generation is already feasible if embedding models are chosen carefully. The complete benchmark, data, code, and raw outputs are available at <https://github.com/RameshKadariya/nepali-llm-benchmark>.

Index Terms—low-resource languages, Nepali, large language models, benchmark, tokenization, multilingual NLP, embeddings, retrieval-augmented generation

I. INTRODUCTION

Large language models have become general-purpose infrastructure, but their training data is overwhelmingly English. For the estimated hundreds of millions of people whose primary language is a low-resource one, two practical questions follow. First, how much of a model’s measured capability survives when the interaction language changes? Second, what

hidden costs, in money and in context capacity, do speakers of these languages pay when using commercial systems billed per token?

This paper answers both questions for Nepali (ISO 639-1: *ne*), the official language of Nepal, written in the Devanagari script and spoken by more than 30 million people worldwide. Despite this population, Nepali is firmly a low-resource language in the NLP sense: it constitutes a negligible share of common crawled corpora, has few curated evaluation datasets, and is rarely reported in model cards.

The contributions of this work are:

- 1) A frozen, openly licensed, fully reproducible five-task evaluation suite for Nepali covering tokenizer fertility, reading comprehension, sentiment classification, machine translation, and embedding retrieval, with all sampling fixed by seed and documented provenance.
- 2) A paired experimental design for comprehension: the same 200 questions are evaluated in English and Nepali for every model, so language effects are measured within-model on identical content and tested with McNemar’s exact test.
- 3) Empirical results for six LLMs spanning three orders of magnitude in parameter count (0.5B to 120B plus a frontier commercial model) and six embedding models, all obtained with zero inference budget using local CPU inference and free API tiers, demonstrating that rigorous low-resource evaluation is accessible to independent researchers.
- 4) Quantification of the tokenizer cost penalty for Nepali across twelve tokenizers, connecting tokenizer design directly to the price and context capacity experienced by Nepali users.

All code, frozen data, raw per-item model outputs, and generated figures are public in the project repository.¹

¹<https://github.com/RameshKadariya/nepali-llm-benchmark>

II. RELATED WORK

Multilingual benchmarks. Belebele [1] provides parallel multiple-choice reading comprehension over FLORES passages [2] in 122 language variants, including Nepali, and is the backbone of the comprehension and retrieval tasks here. The FLORES family [2], [3] established parallel evaluation for translation in low-resource settings.

Tokenizer fairness. Petrov et al. [4] showed that subword tokenizers induce large cross-language differences in sequence length, and Ahia et al. [5] connected these differences to real monetary cost in commercial APIs. This work extends that line with current-generation tokenizers (GPT-4o, Llama 3, Qwen 2.5, DeepSeek V3, Gemma 2) measured specifically on Nepali parallel text.

Multilingual embeddings. Multilingual E5 [6], BGE-M3 [7], and LaBSE [8] are widely used for cross-lingual retrieval; Sentence-BERT [9] popularized the sentence embedding paradigm. Their behavior on Nepali retrieval has not previously been reported systematically.

Nepali NLP. Dedicated Nepali resources remain scarce. NepBERTa [10] is a monolingual BERT for Nepali and is included here as a tokenizer reference point. Public Nepali sentiment corpora exist mainly as social media collections; one such corpus is used for the sentiment task with documented cleaning and label verification.

III. BENCHMARK DESIGN

A. Data Sources and Provenance

Two openly licensed sources supply all evaluation data. The exact construction of every frozen file is documented in the repository (`data/PROVENANCE.md`) and produced by a single deterministic script with random seed 42.

Belebele [1] (CC-BY-SA 4.0): the `npi_Deva` and `eng_Latn` test splits contain 900 multiple-choice questions each, aligned across languages by passage link and question number, so every Nepali item has an exact English counterpart over the same content. The passages originate from FLORES-200. Because FLORES+ and IN22 are gated on the HuggingFace Hub, the parallel Nepali-English passage pairs used for the translation and retrieval tasks are extracted from Belebele, which redistributes them under an open license; translation is therefore evaluated at passage level (typically two to five sentences).

NepaliSentiment (HuggingFace: `Shushant/NepaliSentiment`): 6,000 items of authentic Nepali user-generated social media text, primarily stock market and video platform comments. Label semantics are undocumented upstream; they were determined by systematic manual inspection of samples per class (0 = negative, 1 = positive, 2 = neutral). Cleaning removed malformed labels (8 rows), texts shorter than 15 or longer than 500 characters, texts with fewer than 10 Devanagari characters, and exact duplicates, leaving class pools of 1,299 negative, 1,701 positive, and 853 neutral items.

B. Frozen Evaluation Sets

From these sources, the following sets were frozen: (i) 200 paired reading comprehension items per language; (ii) 150 parallel passages for translation; (iii) a retrieval corpus of all 488 unique Nepali passages with 200 queries (one question per passage, gold passage known, with both Nepali and English query variants); and (iv) 210 sentiment items sampled 70 per class. A 30-item validation sample (15 sentiment, 10 comprehension, 5 translation) is included in the repository for native speaker review; the author is a native Nepali speaker.

C. Tasks and Metrics

Task 1, tokenizer fertility: each of twelve tokenizers encodes the 150 parallel passages in both languages. Reported are fertility (tokens per whitespace word) and the cost ratio, defined as Nepali tokens divided by English tokens for the same content.

Task 2, reading comprehension: accuracy on the paired Belebele items. API-served chat models answer generatively at temperature 0 and must reply with the option number; the first digit 1 to 4 in the reply is the prediction and unparseable replies count as incorrect abstentions. The locally run small model is scored by length-normalized log-likelihood over the four options, the standard protocol of the `lm-evaluation-harness`. The within-model language effect is tested with McNemar’s exact test on paired correctness.

Task 3, sentiment: three-class classification of the 210 native items with a Nepali instruction requesting a one-word answer. Accuracy and macro-F1 are reported; chance accuracy is 33.3%.

Task 4, translation: passage-level translation in both directions at temperature 0. `chrF++` [11] is the primary metric because it is robust for Devanagari; BLEU [12] is computed with `sacreBLEU` [13] using `flores200` sentencepiece tokenization when the output language is Nepali.

Task 5, retrieval: each embedding model encodes the 488-passage Nepali corpus and the 200 queries in two settings: monolingual (Nepali query) and cross-lingual (English query). `Recall@{1,5,10}` and `MRR@10` are reported with cosine similarity over normalized embeddings. An English-only model, `all-MiniLM-L6-v2`, is included deliberately as a negative control.

IV. EXPERIMENTAL SETUP

A. Models

Six LLMs are evaluated: `Qwen2.5-0.5B-Instruct` (run locally on a single CPU core in `float32`), `Llama 3.1 8B`, `Llama 4 Scout 17B`, `Llama 3.3 70B`, `GPT-OSS 120B` (all via the Groq free API tier), and `Gemini 3.1 Flash Lite` (via the Google AI Studio free tier). Six embedding models are evaluated locally on CPU: `multilingual-e5-small` and `base` [6], `paraphrase-multilingual-MiniLM-L12-v2`, `LaBSE` [8], `BGE-M3` [7], and `all-MiniLM-L6-v2` as the English-only control. Twelve tokenizers are analyzed, listed in Table I.

TABLE I
TOKENIZER FERTILITY ON 150 PARALLEL PASSAGES. RATIO IS NEPALI
TOKENS DIVIDED BY ENGLISH TOKENS FOR IDENTICAL CONTENT.

Tokenizer	Fert. ne	Fert. en	Ratio ne/en
GPT-2 (r50k)	10.92	1.23	7.59
GPT-4 (cl100k)	6.91	1.24	4.79
Qwen 2.5	6.53	1.26	4.46
Mistral v0.3	7.09	1.38	4.43
DeepSeek V3	4.29	1.23	2.99
Llama 3.1	3.75	1.24	2.60
Gemma 2	3.09	1.24	2.14
GPT-4o (o200k)	2.32	1.23	1.62
mT5	2.64	1.55	1.46
NLLB-200	1.95	1.40	1.19
BLOOM	1.73	1.24	1.18
NepBERTa	2.21	4.09	0.47

B. Zero-Budget Constraint and Partial Samples

The entire study was executed without paid inference, by design, to demonstrate feasibility for independent researchers in low-resource settings. Free API tiers impose daily token quotas; where a quota was exhausted before a run completed, evaluation stopped and the completed sample size is reported explicitly. This affects Llama 3.3 70B comprehension (178 of 200 paired items), GPT-OSS 120B sentiment (158 of 210), Gemini 3.1 Flash Lite sentiment (90 of 210), and the local Qwen2.5-0.5B Nepali run (37 paired items). Translation results for GPT-OSS 120B and Gemini 3.1 Flash Lite were excluded entirely because fewer than ten passages completed. All partial runs are prefix samples of the same fixed random order, not selectively chosen items, and every per-item output is published.

V. RESULTS

A. Task 1: The Tokenizer Cost Penalty

Table I and Fig. 1 report fertility and cost ratios. Three regimes are visible. English-centric tokenizers of the GPT-4 era and popular open models charge Nepali heavily: cl100k at 4.79x, Qwen 2.5 at 4.46x, Mistral at 4.43x. A middle group has partially multilingual vocabularies: DeepSeek V3 at 2.99x, Llama 3.1 at 2.60x, Gemma 2 at 2.14x. Multilingual-first designs approach parity: GPT-4o at 1.62x, mT5 at 1.46x, NLLB-200 at 1.19x, BLOOM at 1.18x. The Nepali-specific NepBERTa inverts the relationship at 0.47x. The practical meaning is direct: a Nepali user of a cl100k-tokenized API pays nearly five times the price of an English user for the same information and fits nearly five times less content into the same context window. The improvement from cl100k to o200k (4.79x to 1.62x) shows the penalty is a vocabulary design choice, not an inherent property of Devanagari.

B. Task 2: Reading Comprehension Collapses, Then Recovers with Scale

Table II and Fig. 2 report paired accuracy. Every model loses accuracy when the identical questions are asked in Nepali, and every gap except the small-sample local model is highly

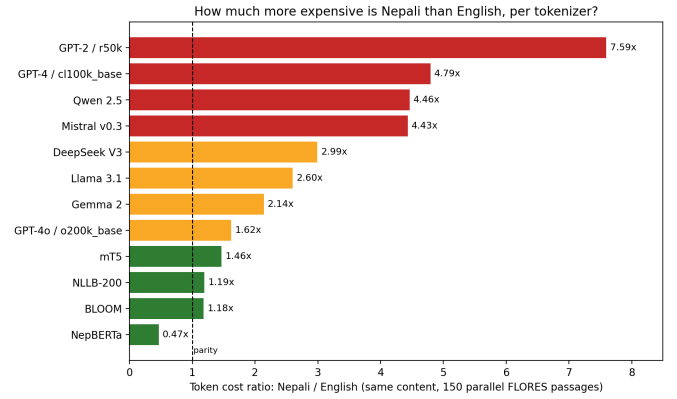


Fig. 1. Token cost ratio (Nepali/English) per tokenizer on identical parallel content.

TABLE II
PAIRED READING COMPREHENSION (BELEBELE). ACCURACY ON ITEMS
COMPLETED IN BOTH LANGUAGES; MCNEMAR EXACT TEST ON PAIRED
CORRECTNESS.

Model	n	Acc en	Acc ne	p
Qwen2.5-0.5B (local)	37	46.0	29.7	0.18
Llama 3.1 8B	200	84.0	40.5	2.7×10^{-23}
Llama 4 Scout 17B	200	96.0	84.5	5.7×10^{-6}
Llama 3.3 70B	178	97.2	84.3	2.4×10^{-7}
GPT-OSS 120B	200	95.0	86.5	4.9×10^{-4}
Gemini 3.1 Flash Lite	200	97.0	89.5	6.1×10^{-5}

significant under McNemar’s exact test. The size of the loss is strongly model-dependent. Llama 3.1 8B collapses from 84.0% to 40.5%, barely above the 25% chance floor, and fails to produce a parseable answer on 71 of 200 Nepali items, indicating that it often cannot even follow the Nepali instruction format. Frontier-class and very large open models retain most of their ability: GPT-OSS 120B drops 8.5 points and Gemini 3.1 Flash Lite drops 7.5 points. The discordant-pair structure is one-sided; for Gemini, 15 items flipped from correct in English to wrong in Nepali and zero flipped the other way.

C. Task 3: Informal Native Nepali Is Much Harder than Formal Nepali

Table III reports sentiment results. Although the same frontier models exceed 86% on formal Nepali comprehension, none exceeds 67.1% accuracy on three-class sentiment over informal native social media text (chance 33.3%). GPT-OSS 120B leads at 67.1% accuracy and 0.650 macro-F1; Llama 3.1 8B manages only 51.9%. Inspection of per-class scores in the raw outputs shows the neutral class is the main failure mode, with models over-predicting emotional labels for factual market commentary. The gap between Belebele performance and this task indicates that formal translated text substantially overstates practical Nepali capability on real user language, which features colloquial spelling, code-mixing, and domain slang.

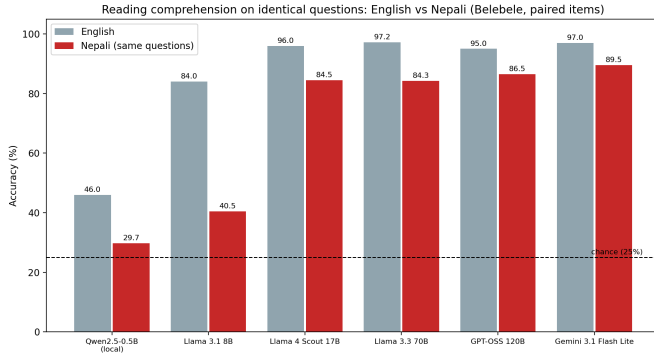


Fig. 2. Reading comprehension accuracy on identical paired questions in English and Nepali.

TABLE III
THREE-CLASS SENTIMENT CLASSIFICATION OF NATIVE NEPALI SOCIAL MEDIA TEXT. CHANCE ACCURACY IS 33.3%. N VARIES WHERE FREE-TIER QUOTAS ENDED RUNS EARLY.

Model	n	Accuracy	Macro F1
Llama 3.1 8B	210	51.9	0.458
Llama 4 Scout 17B	210	58.6	0.484
Gemini 3.1 Flash Lite	90	61.1	0.558
GPT-OSS 120B	158	67.1	0.650

D. Task 4: Models Read Nepali Better than They Write It

Table IV reports translation quality. Both evaluated models are markedly better translating out of Nepali than into it. For Llama 3.1 8B the asymmetry is severe: 53.9 chrF++ from Nepali to English but 30.6 into Nepali, with BLEU collapsing from 28.9 to 7.7. Llama 4 Scout 17B shows the same direction of asymmetry at a higher level (65.5 versus 49.1 chrF++). This mirrors the comprehension results: exposure to Nepali during pretraining appears sufficient for understanding well before it is sufficient for fluent generation.

E. Task 5: Nepali RAG Is Feasible Today, If the Embedding Model Is Chosen Carefully

Table V and Fig. 3 report retrieval quality. BGE-M3 leads with 84.0% Recall@1 and 0.879 MRR@10 monolingually; the multilingual E5 models follow at around 77%. Two failure patterns matter for practitioners. First, the English-only control retrieves essentially nothing (1.0% Recall@1), confirming that default English embedding stacks cannot simply be pointed at Devanagari content. Second, paraphrase-multilingual-MiniLM, a model widely used for multilingual applications, scores only 15.0% monolingually while reaching 52.5% cross-lingually, revealing that its Nepali text understanding, rather than its cross-lingual alignment, is the weak component. Notably, for the strongest models cross-lingual retrieval with English queries performs on par with monolingual retrieval, so English-query-over-Nepali-corpus RAG architectures are viable.

TABLE IV
PASSAGE-LEVEL TRANSLATION QUALITY (150 PARALLEL FLORES PASSAGES). CHRf++ IS PRIMARY; BLEU INTO NEPALI USES FLORES200 TOKENIZATION.

Model	ne → en		en → ne	
	chrF++	BLEU	chrF++	BLEU
Llama 3.1 8B	53.9	28.9	30.6	7.7
Llama 4 Scout 17B	65.5	40.5	49.1	27.3

TABLE V
RETRIEVAL OVER A 488-PASSAGE NEPALI CORPUS, 200 QUERIES. MONO: NEPALI QUERIES. CROSS: ENGLISH QUERIES.

Model	Mono		Cross	
	R@1	MRR@10	R@1	MRR@10
BGE-M3	84.0	0.879	81.0	0.863
mult.-e5-small	77.5	0.829	79.0	0.855
mult.-e5-base	77.0	0.817	76.5	0.829
LaBSE	58.0	0.659	63.5	0.717
para.-mult.-MiniLM	15.0	0.212	52.5	0.619
all-MiniLM-L6 (control)	1.0	0.013	4.5	0.053

VI. DISCUSSION

The Nepali tax is real and layered. A Nepali speaker interacting with a cl100k-era commercial model pays three compounding penalties: a monetary penalty of up to 4.8x per unit of content, a context penalty that shrinks the effective window by the same factor, and an accuracy penalty documented in Tasks 2 to 4. The tokenizer component is fixable by vocabulary design, as GPT-4o demonstrates, and multilingual-first tokenizers show near-parity is achievable.

Scale and recency close the comprehension gap but not the production gap. The paired design isolates language as the only variable, and the ordering is unambiguous: the English-to-Nepali drop shrinks from 43.5 points (Llama 3.1 8B) to under 9 points for the largest and most recent systems. However, generation into Nepali lags comprehension across the board, and informal register remains difficult even for the strongest models tested. Benchmarks built solely on formal translated text, including Belebele itself, will overstate real-world Nepali capability.

Practical guidance. For Nepali RAG systems today: use BGE-M3 or multilingual E5 for retrieval, never English-only defaults; English queries over Nepali corpora work well with these models. For generation, small open models (8B and below) should be assumed unreliable in Nepali until verified; model families differ far more in Nepali than their English leaderboard positions suggest.

Zero-budget reproducibility. Every number in this paper was produced on one CPU core plus free API tiers, with quota-truncated sample sizes disclosed rather than hidden. This is offered as a template for benchmark work on other low-resource languages, where the researchers best positioned linguistically are often the least resourced financially.

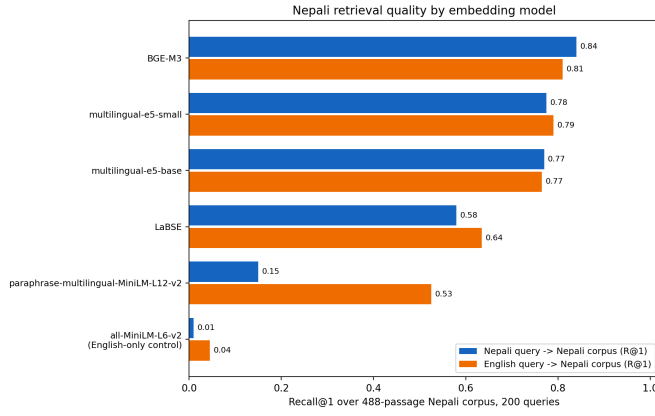


Fig. 3. Recall@1 by embedding model, monolingual and cross-lingual settings.

VII. LIMITATIONS

Sample sizes are modest (150 to 210 items per task) and several API runs are quota-truncated prefix samples, with 90 items for one sentiment model; the affected sample sizes are stated wherever results are reported. Translation is evaluated at passage level rather than sentence level because sentence-aligned open Nepali test sets are gated. A single prompt template per task is used at temperature 0, so prompt sensitivity is unmeasured. Belebele passages are public data and partial training contamination cannot be excluded, though contamination would compress rather than create the observed language gaps, since both language versions are equally public. The sentiment corpus has undocumented upstream labeling methodology; label semantics were verified by manual native speaker inspection, and a validation sample ships with the repository, but the original annotation quality is inherited. Results for gated models such as GPT-4o and Claude could not be included without paid access, and the Gemini flagship quota permitted only a 21-item English probe, which is excluded from all tables. Finally, retrieval uses questions as queries over the passages they were written against, which is a favorable setting compared to open-web Nepali retrieval.

VIII. CONCLUSION

This paper introduced a reproducible, openly licensed, zero-budget benchmark of Nepali capability across five tasks and eighteen models and tokenizers. Current AI systems charge Nepali speakers more per unit of meaning, understand them less reliably, and write their language worse than they read it, but both the cost gap and the comprehension gap are demonstrably closable: modern multilingual tokenizers approach parity, frontier models hold most of their accuracy, and off-the-shelf multilingual embeddings already support practical Nepali retrieval. The benchmark is released in full, including raw per-item outputs, so that future models can be measured against these baselines and so that the same methodology can be applied to other low-resource languages.

DATA AND CODE AVAILABILITY

All code, frozen evaluation data, provenance documentation, raw per-item model outputs, and figure generation scripts are available at <https://github.com/RameshKadariya/nepali-llm-benchmark> under the MIT license, with Belebele-derived data remaining CC-BY-SA 4.0.

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